

PARTS LIST

Semi-conductors:

- IC1 - MC34063A DC to DC converter
 LED1-LED6 - 5mm LED bright white LED
 D1-D3 - SR206 Schottky diode
 Resistors (all 1/4-watt, $\pm 5\%$ carbon):
 R1, R2 - 47-ohm
 R3 - 220-ohm
 R4 - 0.22-ohm
 R5 - 100-ohm
 VR1 - 10-kilo-ohm potmeter

Capacitors:

- C1 - 0.001 μ F ceramic disk
 C2 - 10 μ F, 35V electrolytic

Miscellaneous:

- S1 - SPDT switch
 L1 - 150uH inductor
 CON1 - 2-pin connector terminal
 BATT.1 - 1.2V*two small Ni-MH cells
 - 6V, 100mA solar panel

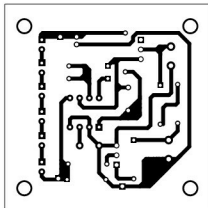


Fig. 3: Actual-size PCB of the kernel light

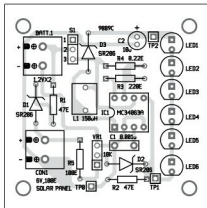


Fig. 4: Component layout of the PCB

proper working before using the circuit.

Even though the datasheet indicates that working voltage of MC34063A is in 3V to 40V range, the IC will start working from 2.4V onwards, but with somewhat lesser efficiency.

For good performance, you must use a 3V (or around 3.6V) battery pack in place of the normal 2.4V type.

Potmeter VR1 is used for automation. If you do not like the automatic switching option, remove components R5, VR1 and D2 from the PCB circuit. ●



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Test Points

Test point	Details
TP0	0V (GND)
Across TP0 and TP1	Around 1.0V
Across TP1 and TP2	Around 18V
Across TP0 and TP2	Around 19V

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